

EEG Feature Selection in Emotion Recognition Using a Fuzzy Information-Theoretic Based Optimization Approach

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Abstract—For electroencephalogram (EEG)-based emotion recognition, various EEG features are extracted from frequency, time, and time-frequency domains for modeling. Nevertheless, there is no a standard subset of EEG features widely accepted in this research field, giving rise to the challenge of curse dimensionality. To cope with the challenge, many EEG feature selection (FS) methods have been put forward based on information theory. Generally, these methods suffer from the issue of delivering a suboptimal result with heuristic search, and they are also inefficient in balancing the influence of different terms like feature relevance and feature redundancy. Based on this, we present a new fuzzy information-theoretic based optimization approach to attain the goal. To be specific, fuzzy mutual information is unitized to evaluate EEG features from the relevance and redundancy perspective, and data structure information is captured to exploit feature manifold simultaneously. Then, a unified optimization framework is designed to take all of them into consideration, thereby inducing a globally optimal result of EEG FS. Extensive empirical studies on three EEG emotional datasets reveal that our method is able to found out a discriminative and nonredundant feature subset from different domains, and therefore achieves the performance improvement of emotion recognition.

Index Terms—Electroencephalogram (EEG), emotion recognition, feature selection (FS), fuzzy information-theoretic, optimization.

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I. INTRODUCTION

EMOTIONS are described as psychophysiological mechanism provoked by the neuronal process of the organisms, and the related research is popular in psychology and neuroscience [1], [2]. Traditionally, emotion recognition studies primarily focus on analyzing speech, facial expressions, or a combination of both. [3], [4], [5]. These modalities induce nonphysiological signal response, and may be unable to reveal the inherent states of the human emotions directly. Instead, electroencephalogram (EEG) is one of physiological signals that is commonly utilized to represent the emotional states. EEG signals are recorded with a noninvasive method, i.e., they are generated by a numerous of EEG electrodes positioned over the scalp. Considering that EEG signals are acquired from the surface of the brain, these are deemed to be strong indicators of emotions, and proved to be effective in providing the discriminative information [6]. Based on this, there is a growing interest in emotion recognition using EEG in recent years [7], [8], [9], [10].

For building such a model using EEG signals, high-dimensional EEG feature vectors are often extracted from the resource in time [11], [12], frequency [7], [13], and time-frequency domains [14], [15]. The extracted data with high dimensionality generally involve lots of irrelevant, redundant, and noisy EEG features. For example, the adjacent electrodes probably provide similar or repeated information, redundant features may be existed in the EEG data [16], [17]. Therefore, it is necessary to identify informative EEG features to solve the high-dimensionality issue, thus reducing the risk of overfitting in recognizing emotion results. Actually, emotional controllability varies with person, resulting in distinct neurophysiological responses to same stimuli. Even with the same cognitive environment, the responses can vary with different persons [6]. By exploiting the EEG features highly correlated with the emotional states, an emotion recognition model can be produced to improve the robust capability, and even enhance the model interpretability.

To address the aforementioned challenge, the application of feature selection (FS) has emerged as a main focus [1], [17], [18], [19], [20]. Among the existing FS methods, many algorithms are based on information theory [21]. These algorithms take the aim at designing feature evaluation criteria, such as feature redundancy minimization [18], [22] and feature

relevance maximization [20], [23], to access the importance of EEG features. For implementation, they often employ a heuristic search strategy to evaluate candidate features one by one, until a size-fixed subset of relevant EEG features is generated for emotion recognition. Nevertheless, these methods with heuristic search are easily trapped in local optima, in the meanwhile, generally they lack the mechanism to balance the influence of involved terms like the relevance and the redundancy. Not surprisingly, it may be difficult to obtain an optimal feature subset, limiting the effectiveness in EEG studies. Furthermore, the current information-theoretic based EEG FS algorithms can only be applied to discrete variables. For continuous values of EEG features, the technique of data discretization [21] has to be used beforehand. Therefore, they may not be able to precisely calculate the information quantity of each feature for EEG FS.

As the above stated, we introduce a novel Fuzzy Information-Theoretic based Optimization method, namely FITO, to conduct EEG FS in emotion recognition. To be specific, we first leverage fuzzy mutual information to estimate the terms of feature relevance and feature redundancy. Then, we construct an optimization framework with the goal of incorporating and trading off them, thus evaluating features from a global view. In consideration of the optimization framework can make the weight of EEG features available, utilizing structure information of EEG feature space becomes convenient to improve the generalization performance. Therefore, we further use a sparse reconstruction method to exploit data manifold among EEG features, thus regularizing the optimization model. Empirical studies on three EEG emotional datasets demonstrate that our proposed FITO method is effective in EEG FS for emotion recognition. What we should note is regarding the model interpretability. Specially, our method is able to locate tailored EEG features from different feature types. Based on the selected EEG features, FITO also can inform important electrodes that help to associate with emotional brain activities.

In summary, our contributions can be concluded as follows.

- 1) To capture the relevance and the redundancy of EEG features, we propose a global optimization approach with fuzzy mutual information, thus achieving discriminative FS for emotion recognition. Moreover, a feature manifold regularizer is incorporated with sparse representation to enhance the generalization performance.
- 2) The analysis of the selected features by the proposed method reveals that discriminative features vary across datasets, emphasizing the necessity of FS. In addition, we consistently observe that the features from AF3 in the prefrontal lobe are selected, indicating that the brain region and the electrode are significant for recognizing the emotion states.
- 3) Experiments are conducted on three EEG emotional datasets, and two widely used metrics are applied to compare, i.e., classification accuracy and representation entropy. The experimental result reveals that our method has the advantages in EEG FS.

The rest of this article is organized as follows. We describe the used symbols and review the related work on EEG FS in Section II. Specially, the research on information-theoretic based

EEG FS methods is discussed in detail. Section III introduces the concepts of fuzzy entropy and fuzzy mutual information. Section IV describes our proposed method, the optimization solution, and some other implementation issues. The experimental results are analyzed in Section V. Finally, Section VI concludes this article.

II. NOTATIONS AND RELATED WORK

A. Notations and Definitions

We define some important symbols used throughout this article as follows. Formally, we denote a vector and a matrix with a lowercase letter and an uppercase letter, respectively. For example, a vector can be denoted with $\mathbf{a}$, $\mathbf{b}$, or $\mathbf{c}$..., and a matrix can be denoted with $\mathbf{A}$, $\mathbf{B}$, or $\mathbf{C}$ The trace of a matrix, e.g., $\mathbf{A}$, is represented by $\text{tr}(\mathbf{A})$, and its transpose is represented by $\mathbf{A}^T$. Let $\{\mathbf{F}, \mathbf{y}\}$ be an EEG emotional dataset, $\mathbf{F} = [\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_d] \in \mathbb{R}^{n \times d}$ denotes the feature matrix extracted from EEG recordings, where $\mathbf{f}_i (1 \leq i \leq d)$ is the vector including the information of n samples *w.r.t.* the i -th feature. In addition, $\mathbf{y} \in \mathbb{R}^{n \times 1}$ is the corresponding ground-truth label information. Then in the dataset $\mathcal{U}$, any sample can be represented with a d -dimensional feature vector, and attributed to a certain class.

B. Related Work

EEG feature selection (FS) aims at removing irrelevant, redundant, even noisy features in EEG data, thus preventing the overfitting of a learning model, and enhancing the model interpretability. Typically, the existing EEG FS methods can generally be categorized into three groups based on their selection strategies, i.e., filter, embedded, and wrapper [21]. Filter methods are developed based on certain data characteristics, and generate a feature ranking solely as the input of a classification model [24]. Different to the former, embedded methods view the FS process as a function component of the predefined classifier [25]. Finally, wrapper methods make an evaluation of selected features based on the performance of an off-the-shelf classifier [13]. Based on the above strategies, there are lots of salient studies specific to EEG FS. For instance, Sun et al. [26] proposed a clustering-fusion FS method on the basis of resting state EEG signals, thus recognizing major depressive disorder. Kannadasan et al. [6] introduced a differential-evolution based EEG FS approach to identify arousal and valence emotional states. By applying different fitness functions and stopping criteria, Saibene and Gasparini [13] proposed to use the adoption of a genetic algorithm based on EEG heterogeneous data. Xu et al. [17] devised to select discriminative EEG features by global redundancy minimization in orthogonal regression. Pippa et al. [25] introduced an automated FS framework with ReliefF and BayesNet, and therefore classified epileptic and nonepileptic EEG events.

For emotion recognition relying on EEG FS, extensive tailored methods have been introduced to serve the community in recent years, specially, information-theoretic based FS methods have received an impressive attention in terms of recognition performance. One representative strategy based on

information theory is to maximize feature relevance, i.e., the shared information of candidate EEG features with emotional label information $\mathbf{y}$. Suppose that $\mathbf{f}^+$ denotes a candidate EEG feature, the framework can be generalized as

$$L_{Max_Rel}(\mathbf{f}^+) = MI(\mathbf{f}^+, \mathbf{y}) \quad (1)$$

where $MI(\mathbf{f}^+, \mathbf{y})$ represents the mutual information between $\mathbf{f}^+$ and $\mathbf{y}$, which measures their shared information [27]. Following this strategy, Wang et al. [23] applied normalized mutual information to the selection of EEG channels. Piho and Tjahjedi [1] introduced a FS method via windowing to search for each reduced EEG signal. Chen et al. [20] considered using the technique (like information gain) to develop an EEG-based emotion assessment system. Although these methods are used to conduct EEG FS well, they may easily lead to an unsatisfactory result while the term of feature redundancy is ignored. For that, some other information-theoretic based EEG FS methods are proposed, which can be mainly divided into two categories, i.e., 1) feature redundancy minimization and 2) newly classification information maximization.

EEG FS via feature redundancy minimization: Methods in this category not only pay attention to the feature relevance with label information, but also require the selected features that are not highly correlated with each other. Based on this, we can formulate the general FS framework, as follows:

$$L_{Red}(\mathbf{f}^+) = MI(\mathbf{f}^+, \mathbf{y}) - MI(\mathbf{f}^+, S) \quad (2)$$

where $MI(\mathbf{f}^+, S)$ represents the redundancy of $\mathbf{f}^+$ while S is known, in which S denotes the subset of selected features. A number of public works leverage this framework to conduct EEG data reduction. For example, Atkinson and Campos [22] incorporated an EEG FS module into the learning framework based on the mRMR criterion, and then designed an adaptive BCI-based mode for emotion recognition. Lin and Tang [28] proposed a new method by using the multiinformation to measure the redundant information between EEG features. Chen et al. [18] first extracted EEG characteristics from data, such as the bicoherence, complexity, and power spectrum. Based on this, they tried to employ mRMR and random probe to achieve discriminative FS for recognizing attention-deficit/hyperactivity disorder.

EEG FS via newly classification information maximization: Methods in this category are based on the assumption that the subset of selected features should be with a small redundancy, simultaneously it is capable to induce more supplementary information for classification. The general framework can be defined as follows:

$$L_{New}(\mathbf{f}^+) = MI(\mathbf{f}^+, \mathbf{y}|S). \quad (3)$$

It is worth noted that $L_{New}(\mathbf{f}^+)$ measures the newly classification information provided by $\mathbf{f}^+$ while the selected feature subset S is given. Based on this setting, some well-established criteria were proposed for feature evaluation, such as joint (or conditional) mutual information maximization [29], [30]. Toward EEG FS, these criteria are commonly used directly, thereby achieving some typical applications like patient-specific seizure detection [31], or others [19].

In addition, deep learning techniques have demonstrated their ability to effectively extract significant features, enabling direct manipulation of raw EEG data without preprocessing or manual feature extraction. There are some methods considering learning feature representation from EEG signals with deep learning. For example, they achieved the purpose based on convolutional neural network [32], [33], [34], long short-term memory [34], or deep belief network (DBN) [35]. These methods enable to facilitate the learning of EEG features, but they are always not intuitional and inexplorable. Moreover, they often involve many hyperparameters that have to be tuned. In the light of that the EEG data are usually source-limited, these method using deep learning may be powerless in recognizing emotion states.

Note that different to the abovementioned information-theoretic based FS algorithms, a novel fuzzy information-theoretic based optimization approach is presented for EEG FS in this article. Specially, we exploit the feature relevance and the feature redundancy by using fuzzy mutual information, and consider to incorporate them, even the information of EEG feature structure, into a unified global optimization framework. Based on this, an EEG FS algorithm is expected to be designed with nice guarantee *w.r.t.* recognition performance. In addition, the proposed method conduct EEG FS by learning the feature weight, therefore, it is with a good interpretability, i.e., our method can inform factors playing important roles for emotion recognition.

III. FUZZY ENTROPY AND FUZZY MUTUAL INFORMATION

For convenience, we describe some fuzzy information-theoretic based measures in this section.

Let $\mathcal{U} = \{x_1, x_2, \dots, x_n\}$ be a nonempty finite set of samples, in which x_i ($1 \leq i \leq n$) is the i th sample and described by a set of fuzzy or real-valued features. We denote the feature set as F , and R is a fuzzy equivalence relation over $\mathcal{U}$ induced by F , the fuzzy relation matrix $\mathbf{M}_R$ can be defined as follows:

$$\mathbf{M}_R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{bmatrix}. \quad (4)$$

Arbitrary element in $\mathbf{M}_R$ reflects the relationship of two samples, e.g., r_{ij} for x_i and x_j , whose value is ranged from 0 to 1. Here, R has the following properties:

- 1) reflexivity: $R(x_i, x_i) = 1 \forall x_i \in \mathcal{U}$;
- 2) symmetry: $R(x_i, x_j) = R(x_j, x_i) \forall x_i, x_j \in \mathcal{U}$;
- 3) transitivity: $R(x_i, x_z) \geq \min_{x_j} \{R(x_i, x_j), R(x_j, x_z)\}$.

Attributed to x_i and the fuzzy set R , a fuzzy equivalence class can be written as follows:

$$[x_i]_R = r_{i1}/x_1 + r_{i2}/x_2 + \cdots + r_{in}/x_n \quad (5)$$

where the symbol “/” means a separator, and the symbol “+” means “union.” Then, we can define the fuzzy cardinality of the fuzzy equivalence class, i.e., $|[x_i]_R| = \sum_{j=1}^n r_{ij}$.

Definition 1 (See [36]): Given an information table $\langle \mathcal{U}, R \rangle$, the fuzzy information entropy of F can be estimated as

$$FE(R) = FE(F) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|[x_i]_R|}{n}. \quad (6)$$

If R is a crisp equivalence relation, and we have

$$R(x_i, x_j) = \begin{cases} 1, & \text{if } x_i \Leftrightarrow x_j \\ 0, & \text{if } x_i \nLeftrightarrow x_j. \end{cases} \quad (7)$$

Then, the fuzzy information entropy can be degenerated into Shannon's information entropy.

Definition 2 (See [36]): Given $\langle \mathcal{U}, R \rangle$, two features $\mathbf{f}_i$ and $\mathbf{f}_j$ are in F , their fuzzy joint entropy is defined as follows:

$$FE(\mathbf{f}_i, \mathbf{f}_j) = -\frac{1}{n} \sum_{t=1}^n \log \frac{|[x_t]_{\mathbf{f}_i} \cap [x_t]_{\mathbf{f}_j}|}{n} \quad (8)$$

where $[x_t]_{\mathbf{f}_i} \cap [x_t]_{\mathbf{f}_j} = \min\{|[x_t]_{\mathbf{f}_i}|, |[x_t]_{\mathbf{f}_j}|\}$.

Definition 3 (See [36]): Given $\langle \mathcal{U}, R \rangle$, two features $\mathbf{f}_i$ and $\mathbf{f}_j$ are in F . The fuzzy conditional entropy of $\mathbf{f}_j$ conditioned to $\mathbf{f}_i$ is computed as

$$FE(\mathbf{f}_j|\mathbf{f}_i) = -\frac{1}{n} \sum_{t=1}^n \log \frac{|[x_t]_{\mathbf{f}_i} \cap [x_t]_{\mathbf{f}_j}|}{|[x_t]_{\mathbf{f}_i}|}. \quad (9)$$

Definition 4 (See [36]): Given $\langle \mathcal{U}, R \rangle$, two features $\mathbf{f}_i$ and $\mathbf{f}_j$ are in F , their fuzzy mutual information can be estimated as

$$FM(\mathbf{f}_i; \mathbf{f}_j) = -\frac{1}{n} \sum_{t=1}^n \log \frac{|[x_t]_{\mathbf{f}_i}| \cdot |[x_t]_{\mathbf{f}_j}|}{n \cdot |[x_t]_{\mathbf{f}_i} \cap [x_t]_{\mathbf{f}_j}|}. \quad (10)$$

Note that the fuzzy mutual information and the fuzzy entropy satisfy the following:

- 1) $FM(\mathbf{f}_i; \mathbf{f}_j) = FE(\mathbf{f}_i) - FE(\mathbf{f}_i|\mathbf{f}_j)$;
- 2) $FM(\mathbf{f}_i; \mathbf{f}_j) = FE(\mathbf{f}_j) - FE(\mathbf{f}_j|\mathbf{f}_i)$;
- 3) $FM(\mathbf{f}_i; \mathbf{f}_j) = FE(\mathbf{f}_i) + FE(\mathbf{f}_j) - FE(\mathbf{f}_i, \mathbf{f}_j)$.

In short, fuzzy mutual information is an effective tool that can be adapted to deal with hybrid data (involving continuous and discrete values). In comparison with the Shannon's mutual information that is only applicable in discrete data, the fuzzy information-theoretic based measure method has the advantage in capturing complex relationships from data.

IV. METHODOLOGY

A. Preliminaries

Toward emotion recognition, a popular and effective way is to associate emotional changes by analyzing EEG signals. In this article, we extract features from EEG signals in frequency, time, and time-frequency domains, and take the aim to search for significant ones, which can be identified and understood, and therefore enhance the recognition performance and interpretability to achieve the purpose. Technically, an EEG FS approach is served with fuzzy information-theoretic based optimization. For leveraging information theory to conduct FS, Bell and Wang [37] have presented the first axiomatic method for reference.

Axiom 1.1 (Preservation of learning information): Given a dataset, the feature space and the label space are labeled as

$\mathbf{F}$ and $\mathbf{y}$, respectively. Suppose that a feature subset $\mathbf{F}_{opt}$ is available, $MI(\mathbf{F}_{opt}, \mathbf{y}) = MI(\mathbf{F}, \mathbf{y})$ in which $MI(\cdot, \cdot)$ is the measure function for mutual information calculation. Then, $\mathbf{F}_{opt}$ is sufficient.

Axiom 1.2 (Minimum encoding length): Given a dataset, the feature space and the label space are labeled as $\mathbf{F}$ and $\mathbf{y}$, respectively. In addition, $\mathbf{F}_{opt}$ is a feature subset. If arbitrary feature $\mathbf{f}_i$ in $\mathbf{F}_{opt}$ can make the joint entropy $H(\mathbf{f}_i, \mathbf{y})$ minimum, $\mathbf{f}_i$ should be favored regarding its ability for prediction.

Axioms 1.1 and 1.2 give an axiomatic description for obtaining an optimal subset of features based on information theory. They are developed based on the principle of Occams razor. In the following of them, there are a number of feature evaluation criteria specific to information-theoretic based FS, involving the relevance, the redundancy, the synergy, and so on [21], [38], [39]. Considering that many refined methods are put forward with feature relevance maximization and feature redundancy minimization (mRMR) ((2) gives the basic FS framework), we inherit the advantage of this criterion to search for important EEG features. In addition, theoretically the measure method of fuzzy mutual information can be adapted for better capturing complex relationships for hybrid EEG data characterized by continuous and discrete information. Thus, we leverage fuzzy mutual information to the estimation of relevance and redundancy *w.r.t.* each EEG feature, along with the mRMR criterion, thereby achieving EEG FS for effective emotion recognition

$$\max_{\mathbf{f}^+} FM(\mathbf{f}^+, \mathbf{y}) - FM(\mathbf{f}^+, \mathcal{S}). \quad (11)$$

In (11), $FM(\cdot, \cdot)$ denotes the measure function for fuzzy mutual information calculation. $FM(\mathbf{f}^+, \mathbf{y})$ evaluates the relevance between candidate EEG feature $\mathbf{f}^+$ and label information $\mathbf{y}$; $FM(\mathbf{f}^+, \mathcal{S})$ estimates the redundant information of $\mathbf{f}^+$ while the selected feature subset $\mathcal{S}$ is given.

B. Fuzzy Information-Theoretic Based Optimization

As shown in (11), EEG FS is conducted using a heuristic method, i.e., an EEG feature subset is heuristically identified with incremental search. As we know, it may be in trouble to evaluate each EEG feature from a global view. Consequentially, a suboptimal FS result is probably delivered for emotion recognition. In addition, although both feature relevance and feature redundancy are taken into consideration in (11), there is the lack of explicit definition and mechanism to trade off the two terms. Many methods consider solving the problem by adjusting the influence of feature redundancy to guide the FS process. For example, Meyer et al. [40] expanded the redundancy item by mining complementary information shared between selected features and unselected features while label information is given. Fleuret [30] achieved the purpose via conditional mutual information maximization. Nevertheless, they may suffer from a lot of complex and ineffective entropy calculations, leading to unreliable evaluation of feature importance. To track these problems, we propose a fuzzy information-theoretic based optimization framework for EEG FS. To be specific, instead of evaluating candidate EEG feature $\mathbf{f}^+$, the criterion, namely (11), can be transformed into the one to generate a feature coefficient vector,

as follows:

$$\max_{\mathbf{w}} \sum_{i=1}^d \left[FM(\mathbf{f}_i, \mathbf{y})w_i - \sum_{j=1}^d FM(\mathbf{f}_i, \mathbf{f}_j)w_iw_j \right] \quad (12)$$

where $\mathbf{w} \in \mathbb{R}^{d \times 1}$ is the feature coefficient vector. Both w_i and w_j are the element in $\mathbf{w}$, which reveal the importance of the i th and j th EEG features, respectively. In (12), we can observe that each EEG feature, e.g., $\mathbf{f}_i$, is evaluated by maximizing the relevance between $\mathbf{f}_i$ and $\mathbf{y}$ (as shown in the first term), whose importance also has negative correlation with the redundancy between the feature and all the other EEG features (as shown in the second term). Following this principle, we can exploit EEG FS globally for emotion recognition by learning the feature weight $\mathbf{w}$. To provide a compact formulation, (12) can be rewritten as the problem of least square programming

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w} - \mathbf{c}\|_F^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{G} \mathbf{w} \quad (13)$$

where α is the tradeoff parameter to control the influence of feature redundancy. In addition, $\mathbf{c}$ is used to preserve the relevance of each EEG feature with label information, and arbitrary element $c_{ij} \in \mathbf{c}$ can be estimated with the fuzzy mutual information $FM(\mathbf{f}_i; \mathbf{y})$, as follows:

$$c_{ij} = -\frac{1}{n} \sum_{t=1}^n \log \frac{|[x_t]_{\mathbf{f}_i}| \cdot |[y_t]_{\mathbf{y}}|}{n \cdot |[x_t]_{\mathbf{f}_i} \cap [y_t]_{\mathbf{y}}|} \quad (14)$$

where y_t is the label information of the t th EEG sample. $\mathbf{G}$ is a symmetric matrix, and used to estimate the redundancy between EEG features. For arbitrary element $g_{ij} \in \mathbf{G}$, $g_{ij} = FM(\mathbf{f}_i; \mathbf{f}_j)$, as follows:

$$g_{ij} = -\frac{1}{n} \sum_{t=1}^n \log \frac{|[x_t]_{\mathbf{f}_i}| \cdot |[x_t]_{\mathbf{f}_j}|}{n \cdot |[x_t]_{\mathbf{f}_i} \cap [x_t]_{\mathbf{f}_j}|} \quad (15)$$

C. Incorporating Feature Manifold

Through optimizing (13), we can achieve EEG FS for emotion recognition while the weight of each EEG feature is available, i.e., the vector $\mathbf{w}$. By virtue of this, our proposal is easily scalable to the geometric structure exploitation of feature manifold. In general, such structural information is deemed to be important in capturing a feature mapping, or a feature coefficient [41], [42]. Based on this, we further construct a regularization item of feature manifold on $\mathbf{w}$, thus enhancing the generalization performance for EEG FS.

To be specific, feature manifold is reconstructed with an *one-to-all* style, i.e., it considers to be recovered by modeling the relationship between one EEG feature and all the other EEG features. For implementation, we serve a scheme by using sparse representation, and exert an empirical function on l_1 -norm regularization term for the reconstruction of feature space, as follows:

$$\min_{\mathbf{b}_i} \|\mathbf{F}_{-i} \mathbf{b}_i - \mathbf{f}_i\|_2^2 + \lambda \|\mathbf{b}_i\|_1 \quad (16)$$

where $\mathbf{F}_{-i} = [\mathbf{f}_1, \dots, \mathbf{f}_{i-1}, \mathbf{f}_{i+1}, \dots, \mathbf{f}_d] \in \mathbb{R}^{n \times (d-1)}$ represents the matrix formed by all EEG features other than $\mathbf{f}_i$, and

$\mathbf{b}_i = [b_{1i}, \dots, b_{i-1,i}, b_{i+1,i}, \dots, b_{di}]^T$ represents the $(d-1)$ -dimensional reconstruction coefficient vector. Besides, λ is a parameter, empirically it is set to $\frac{1}{100} \|\mathbf{f}_i^T \mathbf{F}_{-i}\|_\infty$. After optimizing (16) for each EEG feature, accordingly the weight matrix $\mathbf{B}$, formed by $\{\mathbf{b}_i\}_{i=1}^d$, is obtained with all diagonal elements set to 0. By using the transferred geometric structure, feature manifold can be reconstructed and expected to induce a better result for EEG FS. Therefore, it leads to minimize the following term:

$$\min_{\mathbf{w}} \|\mathbf{w} - \mathbf{B} \mathbf{w}\|_F^2. \quad (17)$$

D. Solution and Implementation Issues

Combining (13) with (17), the final objective function of our method is obtained, as follows:

$$\min_{\mathbf{w}} L(\mathbf{w}) = \frac{1}{2} \|\mathbf{w} - \mathbf{c}\|_F^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{G} \mathbf{w} + \frac{\beta}{2} \|\mathbf{w} - \mathbf{B} \mathbf{w}\|_F^2. \quad (18)$$

In (18), β is the parameter to balance the component in incorporating feature manifold. Note that $\mathbf{c}$, $\mathbf{G}$, and $\mathbf{B}$ are the constant. Among them, $\mathbf{c}$ and $\mathbf{G}$ can be estimated by using fuzzy mutual information, as shown in (14) and (15), respectively. In addition, $\mathbf{B}$ can be available by optimizing (16). In the light of the l_1 -norm regularization is involved, (16) is a convex but nonsmooth optimization problem. For solving this, the alternating direction method of multiplier (ADMM) [43] can be employed to generate the optimal solution. Therefore, instead of optimizing the intractable problem, we can reformulate (16) into the following equivalent form:

$$\min_{\mathbf{b}_i, \mathbf{z}_i, \mathbf{z}_i = \mathbf{b}_i} \|\mathbf{F}_{-i} \mathbf{b}_i - \mathbf{f}_i\|_2^2 + \lambda \|\mathbf{z}_i\|_1. \quad (19)$$

In (19), a new variable $\mathbf{z}_i$ is added, which satisfies the constraint $\mathbf{z}_i = \mathbf{b}_i$. Based on the ADMM procedure, (19) can be solved efficiently while it is transformed into a sequence of unconstrained problems with augmented Lagrangian function, as follows:

$$L_0(\mathbf{b}_i, \mathbf{z}_i, \mathbf{t}_i) = \frac{1}{2} \|\mathbf{F}_{-i} \mathbf{b}_i - \mathbf{f}_i\|_2^2 + \lambda \|\mathbf{z}_i\|_1 + \mathbf{t}_i^T (\mathbf{b}_i - \mathbf{z}_i) + \frac{\rho}{2} \|\mathbf{b}_i - \mathbf{z}_i\|_2^2 \quad (20)$$

where $\mathbf{t}_i$ denotes the estimation of Lagrange multiplier, ρ denotes a penalty parameter. With this, we update $\mathbf{b}_i$, $\mathbf{z}_i$, and $\mathbf{t}_i$ by the scaled ADMM iterations, until obtain the optimal solution for them

$$\begin{cases} \mathbf{b}_i^{(k+1)} = (\mathbf{F}_{-i}^T \mathbf{F}_{-i} + \rho \mathbf{I})^{-1} (\mathbf{F}_{-i}^T \mathbf{f}_i + \rho \mathbf{z}_i^{(k)} - \mathbf{t}_i^{(k)}), \\ \mathbf{z}_i^{(k+1)} = (\mathbf{b}_i^{(k+1)} + \mathbf{t}_i^{(k)} / \rho - \lambda / \rho)_+ \\ \quad - [-(\mathbf{b}_i^{(k+1)} + \mathbf{t}_i^{(k)} / \rho) - \lambda / \rho]_+, \\ \mathbf{t}_i^{(k+1)} = \mathbf{t}_i^{(k)} + \rho (\mathbf{b}_i^{(k+1)} - \mathbf{z}_i^{(k+1)}). \end{cases} \quad (21)$$

Once all the constants $\mathbf{c}$, $\mathbf{G}$, and $\mathbf{B}$ are in preparation, we can adapt the gradient descent-based method to solve the quadratic programming problem directly, i.e., (18). The partial derivative of $\mathbf{w}$ in terms of objective function L can be calculated as follows:

$$\partial L / \partial \mathbf{w} = \mathbf{I} \mathbf{w} + \alpha \mathbf{G} \mathbf{w} + \beta (\mathbf{I} - \mathbf{B})^T (\mathbf{I} - \mathbf{B}) \mathbf{w} - \mathbf{c} \quad (22)$$

Algorithm 1: Procedure of Fuzzy Information-Theoretic based Optimization framework (FITO).

Input: EEG feature matrix $\mathbf{F}$, emotional label matrix $\mathbf{y}$, and parameters α and β ;

Output: Coefficient vector $\mathbf{w}$.

- 1: Estimate feature relevance based on $\mathbf{c}$ by Eq. (14);
- 2: Estimate feature redundancy based on $\mathbf{G}$ by Eq. (15);
- 3: **for** $i = 1$ to d **do**
- 4: Execute ADMM procedure: Optimize $\mathbf{b}_i$ in Eq. (16) for each EEG feature;
- 5: **end for**
- 6: $\mathbf{w} = [\mathbf{I} + \alpha\mathbf{G} + \beta(\mathbf{I} - \mathbf{B})^T(\mathbf{I} - \mathbf{B})]^{-1}\mathbf{c}$;
- 7: **return** $\mathbf{w}$ for EEG feature selection.

where $\mathbf{I}$ is the identity matrix. Then, $\mathbf{w}$ can be solved by setting $\partial L / \partial \mathbf{w} = 0$. Through some algebraic manipulations, we get the closed-form solution *w.r.t.* $\mathbf{w}$, as follows:

$$\mathbf{w} = [\mathbf{I} + \alpha\mathbf{G} + \beta(\mathbf{I} - \mathbf{B})^T(\mathbf{I} - \mathbf{B})]^{-1}\mathbf{c}. \quad (23)$$

Note that the coefficient information in $\mathbf{w}$ is able to indicate the importance of each EEG feature, and therefore we can return top ranked features with the larger weight for EEG FS.

E. Computational Complexity Analysis

Algorithm 1 provides the pseudocode of our proposal FITO. For FITO, the time complexity is dominated by the fuzzy mutual information calculation in terms of EEG features and the label of emotion states (line 1 and line 2), and the feature manifold can be constructed with ADMM efficiently (line 4). To estimate feature relevance, i.e., $\mathbf{c}$, the time cost is $\mathcal{O}(n^2 dk)$, where n , d , and k denote the number of samples, features, and labels, respectively; To estimate feature redundancy, i.e., $\mathbf{G}$, the time cost is $\mathcal{O}(n^2 d^2)$. Thus, the asymptotic computational complexity of FITO is $\mathcal{O}(n^2 dk + n^2 d^2)$.

Compared with most of existing information-theoretic based FS methods with heuristic search (mentioned in Section II), FITO serves as a global optimization framework to select discriminative EEG features, and only needs to go through the relevant and redundant information one time. Thus, it can avoid repetitive entropy calculations for many feature subsets along with the label, thereby obtaining an optimal solution with high efficiency. However, the terms $\mathcal{O}(n^2 dk)$ and $\mathcal{O}(n^2 d^2)$ emerge as the dominant factors and result in the computational burden, especially when large amounts of samples and features are involved. To cope with the issue for some scenarios (e.g., real-time EEG applications), model simplification can be applied before EEG FS, e.g., candidate feature preprocessing and proportional sample sampling [44], [45], making it powerful in obtaining a good performance at a lower time cost.

V. EXPERIMENTS

A. Dataset Description

To validate the proposed method, we select three public EEG emotional datasets which are DEAP, SEED, and AMIGOS. The detail of these datasets is illustrated in Table I.

TABLE I
DETAIL OF DEAP, SEED, AND AMIGOS DATASETS

Dataset	DEAP	SEED	AMIGOS
Subject no.	32	15	27
Channel no.	32	62	14
Session no.	1	3	1
Video no.	40	15	20
Sample no.	1280	675	540
Stimulus materials	music videos	film clips	videos

The EEG data undergo preprocessing with a band-pass filter ranging from 1 to 50 Hz. Subsequently, independent component analysis is applied to eliminate physiological artifacts, including those from electromyogram, electrooculogram, and electrocardiogram signals. For feature extraction, the entire trials are utilized without segmenting them by sliding windows.

According to the existed researches [7], [14], [46], [47], [48], we extract various features from time, frequency and time-frequency domains based on EEG signals. Among them, EEG features from time domain are adopted to identify the characteristics of the time series, including nonstationary index (NSI) [49], higher-order crossing (HOC) [12], spectral entropy [50], Shannon entropy [51], and C0 complexity [52]. To extract the features from frequency domain, EEG recordings are divided into five frequency ranges, which are delta band (1–4 Hz), theta band (4–8 Hz), alpha band (8–13 Hz), beta band (13–30 Hz), and gamma band (30–50 Hz). The frequency domain features are extracted to denote frequency-domain characteristics in the selected frequency bands [53], including differential entropy (DE), absolute power (AP), and absolute power ratio of the beta band to the theta band (AP_β / AP_θ). In addition, given the nonstationary and nonlinear nature of EEG, time-frequency analysis methods can offer additional knowledge about the dynamic variations in EEG signals for emotion recognition [14], [15]. To obtain the time-frequency domain features, we decompose the raw EEG into several intrinsic mode functions, and then calculate the amplitude of the Hilbert transform of each intrinsic mode function (AHTIMF) and the instantaneous phase of the Hilbert transform of each intrinsic mode function (IPHTIMF) [14]. Moreover, spatical features, i.e., differential asymmetry (DASM) [54] and rational asymmetry (RASM) [55], are also computed for the task of emotion recognition. The detailed definition of total 12 types of EEG features can be found in references [12], [14], [50], [51], [52], [53], [54]. For simplicity, we present the dimension of each feature type for the three selected datasets in Table II.

B. Evaluation Metrics

Two measure metrics are applied to the performance evaluation, i.e., classification accuracy and representation entropy.

Classification accuracy: The metric evaluates the rate of EEG samples which are correctly classified, which is most commonly adopted in the literatures of EEG-based emotion recognition [1], [6], [17], [19], [26]. Suppose that r is the correctly classified sample number and n is the total sample number, the accuracy

TABLE II
DIMENSION OF EACH FEATURE TYPE FOR THE THREE DATASETS

Feature type	DEAP	SEED	AMIGOS
NSI	32	62	14
HOC	320	620	140
Spectral entropy	32	62	14
Shannon entropy	32	62	14
C0 complexity	32	62	14
AP	160	310	70
AP_{β}/AP_{θ}	32	62	14
DE	160	310	70
AHTIMF	160	310	70
IPHTIMF	160	310	70
DASM	70	135	35
RASM	70	135	35

is equivalent to that

$$Acc = \frac{r}{n}. \quad (24)$$

Representation entropy: The metric is a measure of the amount of information compression possible by a FS (or feature extraction) method, which equals to the amount of redundancy present in the generated feature representation [56], [57]. Suppose that $\{\lambda_i\}_{i=1}^s$ are the eigenvalues of the $s \times s$ covariance matrix of an EEG feature set with size s , the calculation of representation entropy is as follows:

$$Ent = - \sum_{i=1}^s \tilde{\lambda}_i \log \tilde{\lambda}_i \quad (25)$$

where $\tilde{\lambda}_i = \lambda_i / \sum_{i=1}^s \lambda_i$. For classification accuracy, the larger the value, the better the performance; for representation entropy, the smaller the value, the better the performance.

C. Comparative Studies

Eight well-established feature selection (FS) method are employed to compare with the proposal FITO. All of them are information-theoretic based methods, including seven heuristic methods namely FJMIIV [58], FJMIUR [59], MRI [56], mRMR [22], CIFE [28], JMI [29], and CMIM [31], especially, FJMIIV and FJMIUR are based on fuzzy information measures. Another comparing method is an optimization method called GRM [60]. FJMIIV combines fuzzy entropy with similarity classifier to conduct FS, in which fuzzy entropy is used to measure the uncertainty of features; FJMIUR is a superior FS approach using fuzzy joint mutual information, which formulates the problem via involving feature relevancy, feature complementarity, and feature redundancy; MRI, JMI, and CMIM are three modified methods to maximize newly classification information, in the meanwhile, the redundancy of selected features can be less; mRMR, CIFE, and GRM are based on the max-relevance-min-redundancy criterion. Different with mRMR and CIFE, GRM regards the FS process as an optimization problem.

For the parameters setting, two parameters α and β of our method are tuned in $\{10^{-3}, 10^{-2}, \dots, 10^3\}$, and the parameters of comparing algorithms are set according to the recommendation of the relevant literatures. As the survey on emotion

recognition [7] suggested, the LIBSVM (with linear kernel) [61] is used as the classifier to evaluate the selected EEG features of each method. To efficiently evaluate the effect of EEG FS, we also employ Bayesian logistic regression (BLR) [62] as the classifier. For the parameter-tuning, we tune all the parameters (if any) for all methods and report the best results. In particular, a grid search strategy is adopted to find the optimal parameter, which is determined by making the average classification result (ACR) on the test data smallest. Here, we define the formula for the ACR as follows:

$$ACR(para) = \sum_{i=1}^{30} Acc_i(\mathbf{f}, \mathcal{U}_t) \quad (26)$$

where $para$ denotes the set of algorithmic parameter(s), $\mathbf{f}$ denotes the classifier, i.e., SVM or BLR, and $\mathcal{U}_t$ is the test set. $Acc_i(\mathbf{f}, \mathcal{U}_t)$ outputs the result of classification accuracy while selecting top- i EEG features.

Finally, 70% of the samples are randomly chosen as the training data, while the remaining 30% as the testing data. Note that all samples of one subject are selected either as training set or test set, which will not exist in the training set and the test set simultaneously. By this setting, we verify the generalization performance of all baseline methods. Similarly to the pipeline in [6] and [17], the training and testing data are randomly split 50 times, and the average emotion recognition result is calculated to the guarantee of unbiasedness.

D. Experimental Results Analysis

In this section, we conduct a comparison of our method with eight information-theoretic based FS methods on three EEG emotional datasets, i.e., DEAP, SEED, and AMIGOS. The performance variation on classification accuracy based on SVM is figured out in Fig. 1 by varying the number of selected features. We find from Fig. 1 that on DEAP and AMIGOS, some FS methods (e.g., FITO, FJMIIV, FJMIUR, MRI, and GRM) can obtain the highest accuracy with a few selected EEG features, while on the SEED dataset, the performance of all the methods first has a significant improvement with the increase of selected EEG features, and then becomes stable, or even degrades. Based on this, we conclude that conducting FS can facilitate effective emotion recognition. In addition, it can be found out from Fig. 1 that FITO can perform the best on all the EEG emotional datasets in terms of accuracy. To be specific, we have the following observations.

- 1) mRMR and CIFE has the similar FS criteria with FITO. Specifically, both of them train a model with the max-Relevance-Min-Redundancy to achieve the purpose. Nevertheless, the two comparing methods apply a greedy strategy to select EEG features, and may not provide an optimal output. Besides, both mRMR and CIFE require a discretization step when processing continuous EEG features, which may result in a limitation of the relevance and redundancy undervalued. Thus, mRMR and CIFE exhibit a poor performance on the three EEG emotional datasets. For similar reasons, MRI, JMI, and CMIM, following the

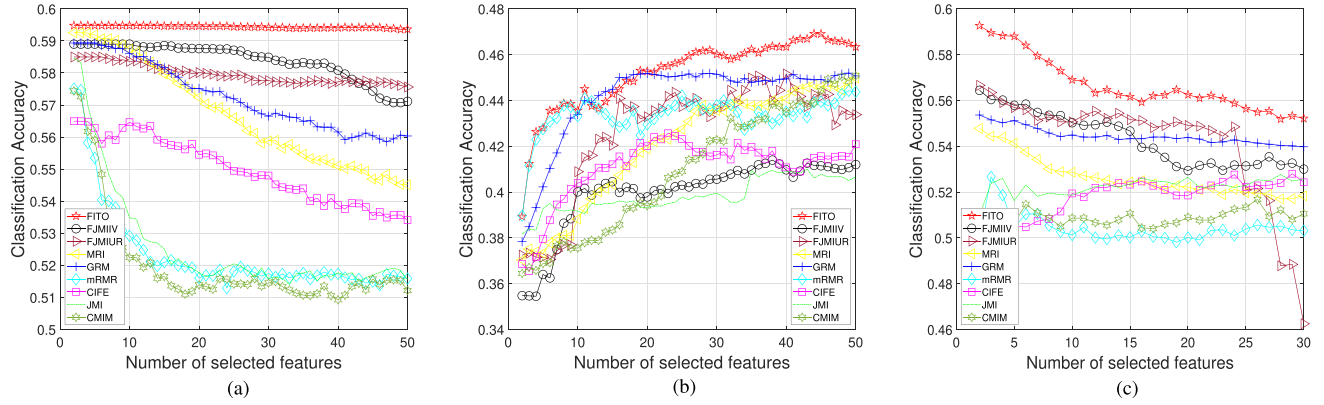


Fig. 1. Variation of accuracy based on selected features by all methods using SVM. (a) DEAP. (b) SEED. (c) AMIGOS.

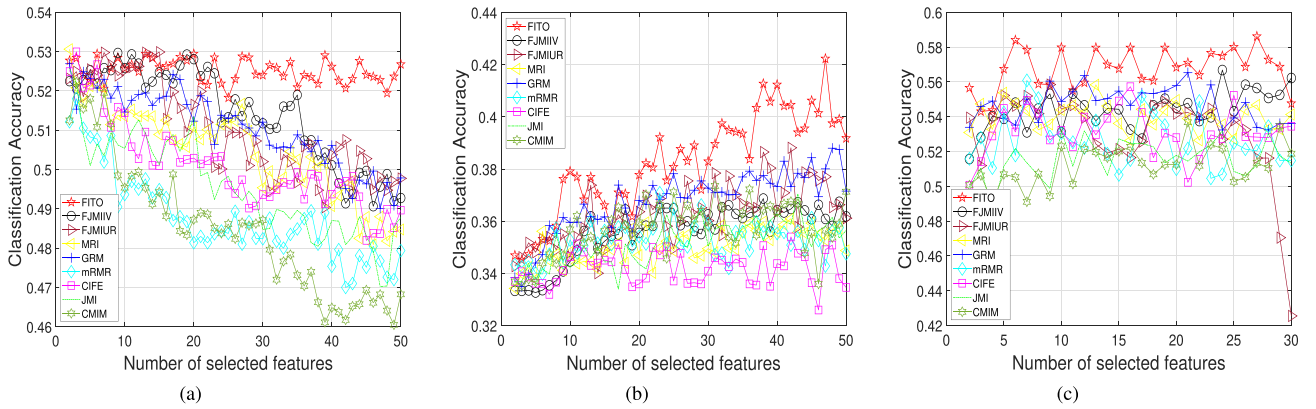


Fig. 2. Variation of accuracy based on selected features by all methods using BLR. (a) DEAP. (b) SEED. (c) AMIGOS.

criteria of newly classification information maximization, also compare unfavorably in the task of EEG FS.

- 2) FJMIUR and FJMIIV are two fuzzy information-theoretic based FS methods. Among them, FJMIUR has considered the relevance and redundancy terms with fuzzy information measures. We can see from Fig. 1 that it can obtain a relatively high accuracy on the three EEG emotional datasets. In addition, FJMIIV employs fuzzy entropy to evaluate the feature dependence, which has a good performance on DEAP, however, fails to get a satisfactory result on SEED. On the whole, the two methods are still inferior to our proposal.
- 3) Similar with our proposal, GRM adopts an optimization strategy to achieve EEG FS by taking feature relevance and feature redundancy in consideration, but it has no the explicit mechanism to balance the influence of feature redundancy, which easily results in misleading selection of EEG features while the relevance and redundancy relations are not estimated accurately. Thus, the proposed FITO method can outperform GRM with respect to accuracy.

Fig. 2 shows the variation of accuracy based on selected features using the classifier BLR. From Fig. 2, we can come to the similar conclusion. Compared with the two fuzzy information-theoretic based methods FJMIUR and FJMIIV, the method

GRM via global redundancy optimization, and the five classical information-theoretic based methods, i.e., MRI, mRMR, CIFE, JMI, and CMIM, FITO still has the advantage in EEG FS.

Stability Analysis: Figs. 1 and 2 demonstrate the feature selection results of all methods regarding classification accuracy. Furthermore, the spider diagram [39] is applied to describe the performance stability of different algorithms on the three datasets based on SVM and BLR. Specially, average classification accuracies of both classifiers across the 50 groups of feature subsets selected by each method are recorded (30 feature groups are generated in AMIGOS).¹ Based on this, the stability index on classification accuracy is displayed in Fig. 3. In Fig. 3, each corner of the spider diagram denotes an EEG emotional dataset exerted by a classifier, and lines with different colors denote different algorithms. When the area enclosed by the same color line is bigger and closer to the regular polygon, the corresponding algorithm of the line has better performance and stability. By observing the result of Fig. 3, we can find out that the shape of FITO verges on a regular tetradecagon, and the area of line closure is the largest among all algorithms. In a nutshell, the stability of our method is obviously optimal.

¹ The first group is composed of the top-1 feature, the second one is composed of the top-2 features, and so on.

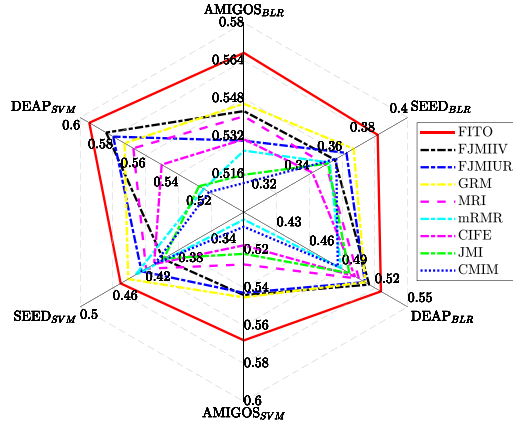


Fig. 3. Stability analysis using spider diagram.

Finally, Fig. 4 shows the performance of all the FS *w.r.t.* representation entropy. From Fig. 4, we can observe that our proposal FITO takes the global redundancy information into consideration with fuzzy mutual information, and can achieve a great improvement in lowering the amount of information compression. Some comparing methods, such as GRM, FJMIUR, and MRI, have their own strength in managing the redundancy of EEG features, but the redundant information cannot be completely eliminated in the selected EEG features. The result in terms of representation entropy further validates that the proposed method is powerful in modeling the dependence among features, thereby retaining remarkable EEG features.

As the above stated, we come to the conclusion that our method is helpful for EEG FS, and can benefit to the performance for emotion recognition.

E. Learning EEG Features: FITO Versus Deep Neural Networks

Both FS and deep learning can be employed to obtain an EEG feature representation. Differently, FS can inform important EEG features, which can be identified and understood, and lead to new insights and hypotheses [63]. However, the features, generated by deep neural networks, may be discriminative to the label of emotion states but lack interpretability.

In this section, we further verify whether the proposed method compares favorably against some state-of-the-art deep neural networks in terms of classification accuracy, and use three deep learning-based methods to compare, namely AE [33], DNN [34], and DBN [35]. Note that the three deep learning-based methods cannot obtain a feature ranking. For validating the proposal with top-ranked EEG features, we employ AE, DNN, and DBN to automatically learn low-dimensional feature representations from the input data. To be specific, 10-D, 30-D, and 50-D feature representations are generated by them on the DEAP and SEED datasets, while on AMIGOS, we use them to generate 10-D, 20-D, and 30-D feature representations for comparison. The classification accuracy, induced by SVM and BLR, based on the selected (or generated) features is shown in Tables III and IV, respectively.

TABLE III
ACCURACY BASED ON SELECTED (GENERATED) FEATURES USING SVM

	# Features	FITO	AE	DNN	DBN
DEAP	10	0.5948	0.5634	0.5221	0.5653
	30	0.5942	0.5535	0.5100	0.5621
	50	0.5936	0.5554	0.5184	0.5600
SEED	10	0.4356	0.4118	0.4565	0.4765
	30	0.4602	0.4319	0.4619	0.4818
	50	0.4635	0.4612	0.4813	0.4961
AMIGOS	10	0.5690	0.5566	0.5309	0.5569
	20	0.5629	0.5569	0.5519	0.5570
	30	0.5521	0.5531	0.5441	0.5502

TABLE IV
ACCURACY BASED ON SELECTED (GENERATED) FEATURES USING BLR

	# Features	FITO	AE	DNN	DBN
DEAP	10	0.5257	0.4900	0.5244	0.4971
	30	0.5240	0.4999	0.5047	0.5059
	50	0.5228	0.4906	0.5065	0.5059
SEED	10	0.3791	0.3692	0.4052	0.3995
	30	0.3829	0.3621	0.4073	0.3984
	50	0.3918	0.3610	0.4125	0.4241
AMIGOS	10	0.5798	0.5097	0.5247	0.4781
	20	0.5684	0.5016	0.5391	0.5225
	30	0.5477	0.5263	0.5441	0.5037

According to Tables III and IV, we can observe that on the DEAP and AMIGOS datasets, FITO achieves the highest classification accuracy with different feature sizes. This suggests that FS is an effective way to seek informative features for EEG-based emotion recognition. On SEED, DNN and DBN perform better than FITO, which both utilize label information to fine-tune their network components (i.e., hidden layers and restricted boltzmann machines), enabling the learning of more discriminative features. In contrast, AE is inferior to our method on this dataset. This is primarily that AE relies solely on self-supervised learning to extract a feature representation from the data, which limits its classification performance. Based on the experimental result on SEED, we infer to that a well-established deep learning-based method may be powerless in expanding interpretability, but can be a better alternative to improve the classification performance.

F. Feature Ranking

In this section, we analyze the importance of features from different domains, and compute the average feature score of each type of features. The result is demonstrated in Fig. 5. Note that each mean feature score is calculated by averaging all the scores of an EEG feature type. For instance, on the SEED dataset, the dimension of the NSI is 62, which is obtained from 62 channels. Therefore, the average feature score for NSI is the mean of the scores of these 62 features.

From Fig. 5, we can observe that the discriminative features are varied on different datasets. For example, the HOC seems to be important for EEG classification on AMIGOS, but useless on the DEAP dataset. Moreover, the DASM seems to be unimportant for EEG classification on DEAP, but crucial on the

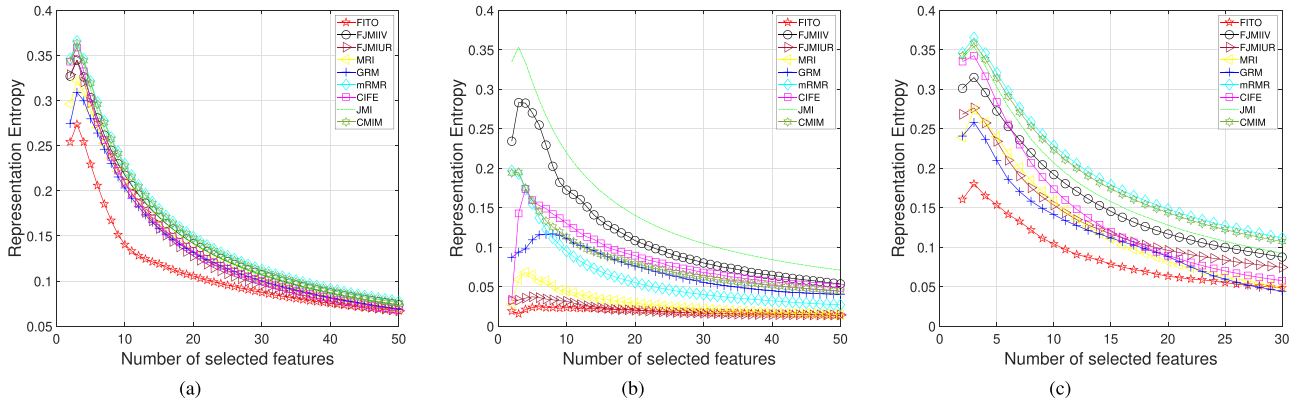


Fig. 4. Variation of representation entropy based on selected features by all methods. (a) DEAP. (b) SEED. (c) AMIGOS.

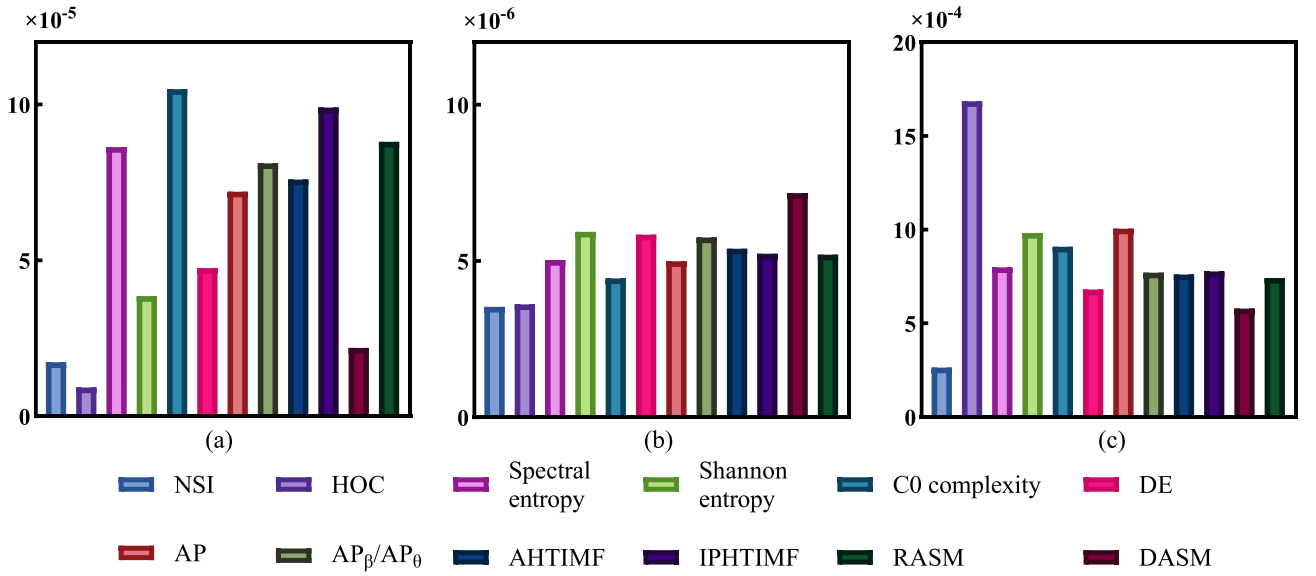


Fig. 5. Importance of various EEG feature categories in emotion recognition, as determined by the feature scores generated by FITO. (a) DEAP. (b) SEED. (c) AMIGOS.

TABLE V
TOP-10 RANKED FEATURES SELECTED BY OUR METHOD ON THE THREE DATASETS

Dataset	1	2	3	4	5	6	7	8	9	10
DEAP	HOC (FC5)	NSI (FC5)	IPHTIMF (F3)	AGTIMF (FC1)	IPHTIMF (F7)	HOC (AF3)	AP_{θ}/AP_{β} (FC1)	AP (AF3)	DE (FC1)	Spectral entropy (FC5)
SEED	HOC (FPZ)	Spectral entropy (AF4)	HOC (AF4)	NSI (AF4)	AGTIMF (F7)	IPHTIMF (F5)	IPHTIMF (AF3)	AP_{θ}/AP_{β} (F7)	HOC (F7)	IPHTIMF (FP2)
AMIGOS	RAMS (AF3)	RAMS (PZ)	RAMS (T7)	DE (T7)	AP_{θ}/AP_{β} (T7)	AGTIMF (F7)	RAMS (FC5)	DE (F8)	RAMS (F3)	AP (T7)

SEED dataset. Based on this, we can draw a conclusion that no one optimal EEG feature type is available that is suitable for all datasets. Therefore, selecting an optimal subset from various feature types is essential to achieve the ideal performance.

For analyzing the performance of FITO, we have compiled the top-10 important features along with their corresponding

electrodes, as exhibited in Tables V. From Table V, we can find out that FITO can select important features from multiple domains, providing rich and low-redundancy information for emotion recognition. In particular, on the DEAP, the subset selected by our proposal includes temporal features (i.e., HOC, NSI, and spectral entropy), frequency features (i.e., DE, AP,

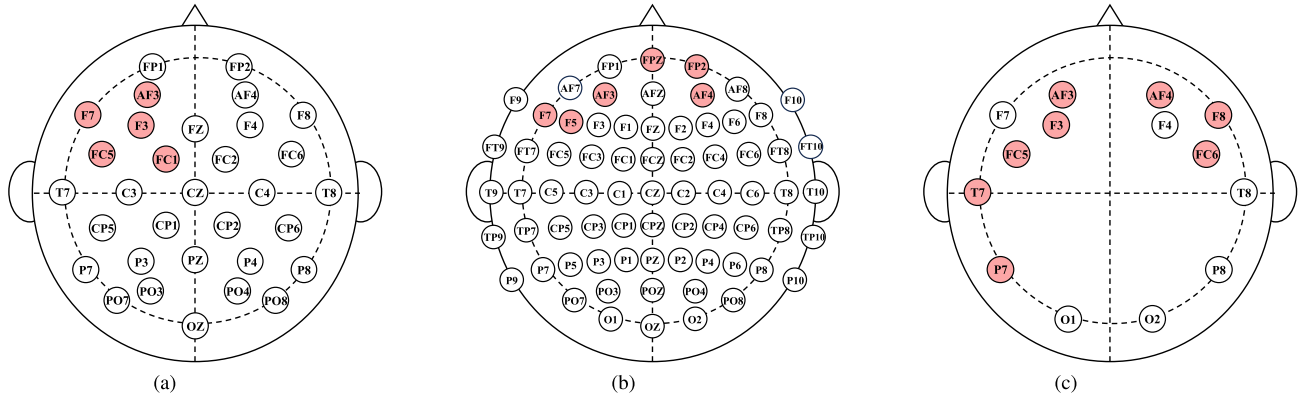


Fig. 6. Distribution of electrodes (corresponding to the top-10 features by using FITO) on the topographical map. (a) DEAP. (b) SEED. (c) AMIGOS.

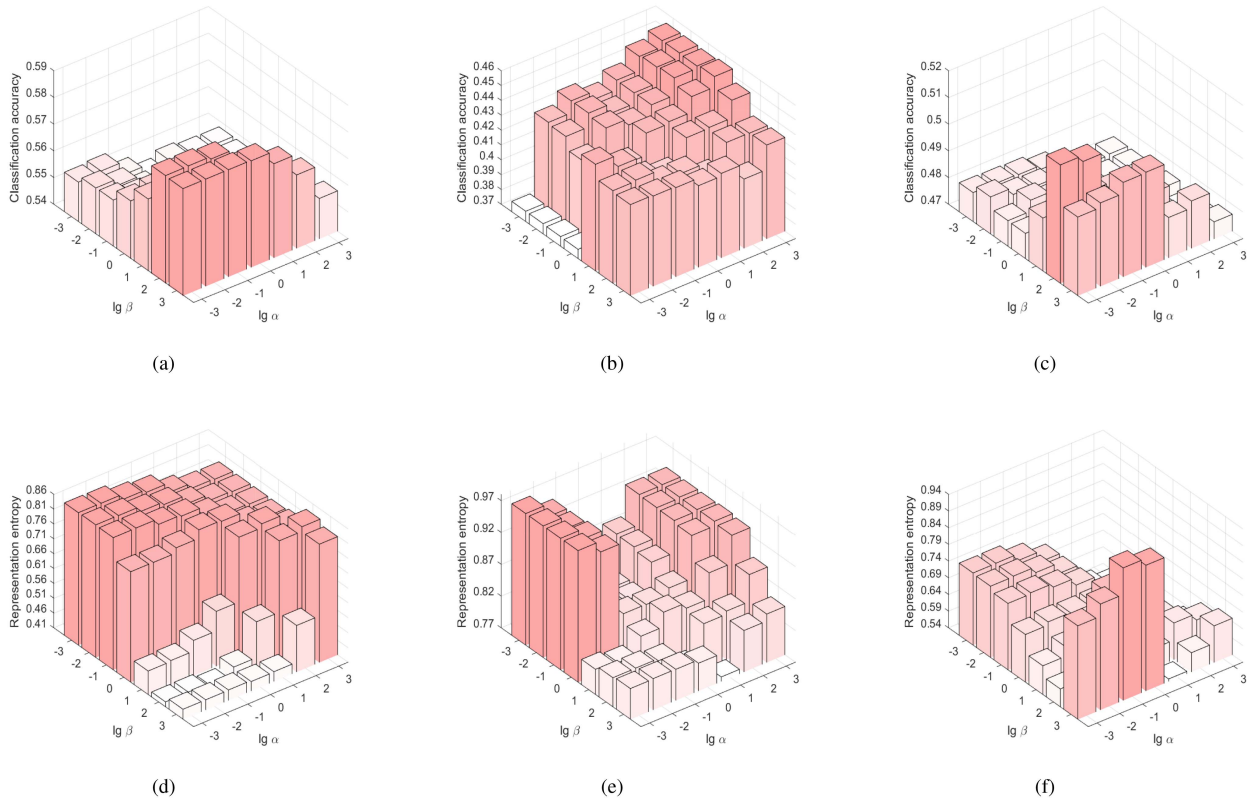


Fig. 7. Parameter sensitivity analysis of FITO. (a) DEAP. (b) SEED. (c) AMIGOS. (d) DEAP. (e) SEED. (f) AMIGOS.

and AP_{θ}/AP_{β}), and time-frequency features (i.e., $IPHTIME$ and $AGTIME$). On the SEED and AMIGOS datasets, a similar phenomenon occurs that EEG features from different domains can be informed for EEG classification. This suggests that the proposed method helps to carry out the most valuable EEG feature subset in different types.

G. Discussion on Selected Electrodes

We further visualize the electrode locations of the selected top-10 EEG features by using our method. Fig. 6 demonstrates the electrode distribution on all the datasets.

Through this experiment, we can explore the brain regions most related to emotional activities by identifying the channels of the important features. In general, the human brain can be categorized into several key regions, including the prefrontal lobe, temporal lobe, parietal lobe, occipital lobe, and others, each playing specific roles in cognitive and emotional functions [64]. Among these regions, the prefrontal lobe (Fp-1/z/2, AF-7/3/z/4/8, and F-7/5/3/1/z/2/3/4/8) has been frequently implicated in the regulation and modulation of emotional states, as corroborated by numerous studies in the literature [65], [66]. Importantly, the electrode AF3 (located at the left anterior frontal area) is often recognized as one of the most significant channels

for reflecting emotional activity in the brain [67], [68], [69], [70]. Based on the analysis of these electrode locations, our method provides insights into the neural circuits most involved in emotional processing, offering valuable information for the development of affective computing systems and emotion-related BCI applications.

According to the experimental results in Fig. 6, we can draw a conclusion consistent with the aforementioned neuro researches [65], [66]. It means the important electrodes are all located at the prefrontal lobe on all the three datasets. Specifically, selected electrodes on all the dataset correspond to regions within the prefrontal lobe, namely in the DEAP dataset, the electrodes are F7, F3, and AF3; In SEED, the electrodes are FPZ, FP2, and AF3; In AMIGOS, the electrodes are AF3, AF4, and F3. This pattern suggests that the features derived from these electrodes effectively capture brain activities related to emotional processing, providing key discriminative information for emotion recognition. Notably, we can observe that the AF3 electrode is selected across all EEG emotional datasets. This consistent selection indicates that the EEG signals from the AF3 electrode are particularly rich in emotion-related information, corroborating findings from neurobiological studies [67], [68], [69], [70]. Summarily, the results of the experiments further provide that the proposed FITO can effectively identify emotion-related features.

H. Sensitivity to Parameters

Our method involves two parameters α and β . Here, we analyze the impact of the two parameters on all the datasets, as demonstrated in Fig. 7.

According to Fig. 7, we first focus on the sensitive of parameter α . Note that α is used to control the influence of feature redundancy, so we analyze α based on the experimental results on the metric of representation entropy, as shown in Fig. 7(d)–(f). From these figures we find out that the low feature redundancy can be obtained by our method while α is set as a relatively small value, or a intermediate value. This is because that the feature redundancy is not fully exploited as it is too small. Once α is large, the term of feature redundancy dominates and leads to the difficulty in generating an optimal EEG feature subset. Thus, the value of α tends to be a compromise. We verify the conclusion by observing the result on classification accuracy, as shown in Fig. 7(a)–(c). It can be seen that following this setting for α , the proposed method can produce a good performance on all the datasets.

We further analyze the sensitive of parameter β . It is set to tradeoff the feature manifold for generalization improvement, hence we focus on the effect of β in terms of classification accuracy, as shown in Fig. 7(a)–(c). We can observe that the proposed method is sensitive to this parameter, and generally a large value of β , e.g., $\beta = 10^2$, helps to the EEG FS.

VI. CONCLUSION

Toward EEG FS in emotion recognition, information-theoretic based methods have received a great attention in past years. However, there exist some shortcomings in the related

algorithms. For example, the selected EEG features are in local optimality. In the meanwhile, they are strongly affected by a relevance evaluation and dependent on each other. Based on this, we propose a new fuzzy information-theoretic based optimization method FITO in this article. The major contribution of our proposal is that it measures feature relevance and feature redundancy with fuzzy mutual information, and then designs a global optimization framework with the relevance and redundancy collaboration to obtain an EEG feature subset. Following this, a promising solution is provided to address the effectiveness problem of EEG FS by employing information theory. Furthermore, it scales well to the problem of leveraging the inherent property of EEG signals. Here, we exploit feature manifold to improve the generalization performance. Experiments on three EEG emotional datasets demonstrate that our method FITO can select a discriminative and nonredundant subset of EEG features. By comparing with eight well-established information-theoretic based FS methods, the advantages of FITO are evaluated in facilitating EEG-based emotion recognition.

In the future, it is interested in further investigating fuzzy mutual information-based optimization approach for EEG FS with multiple feature evaluation criteria. Besides, we have interest in applying the proposed method to some other emotion analysis scenarios, such as mixed emotion exploration [71] and emotion distribution learning [72].

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